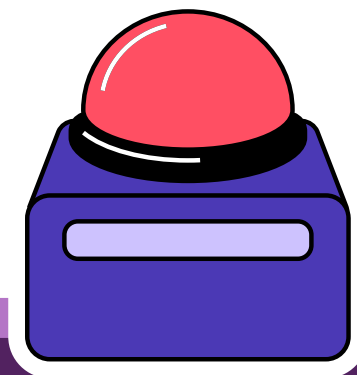
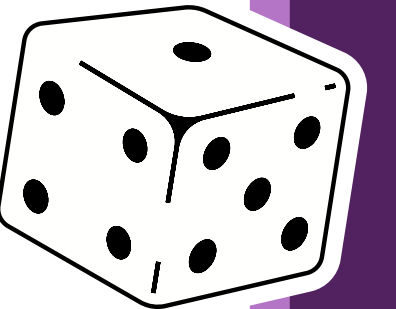


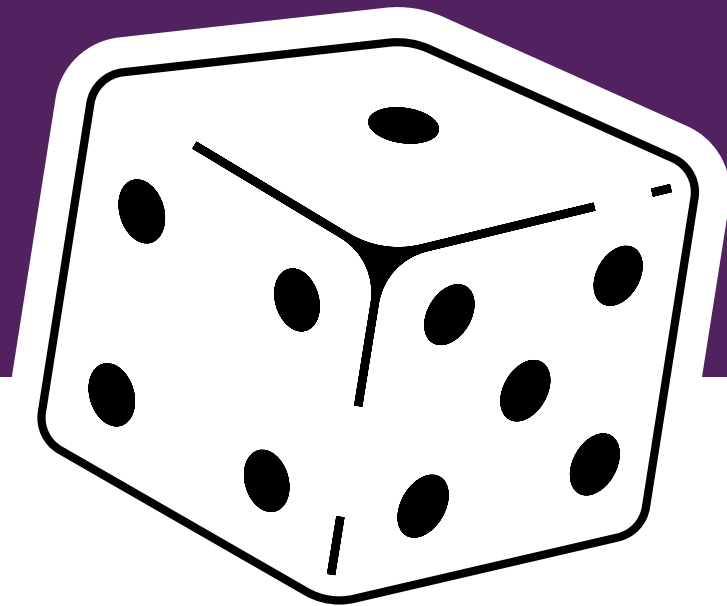
BCC1244 - GAME PROGRAMMING

2D GAME DEVELOPMENT

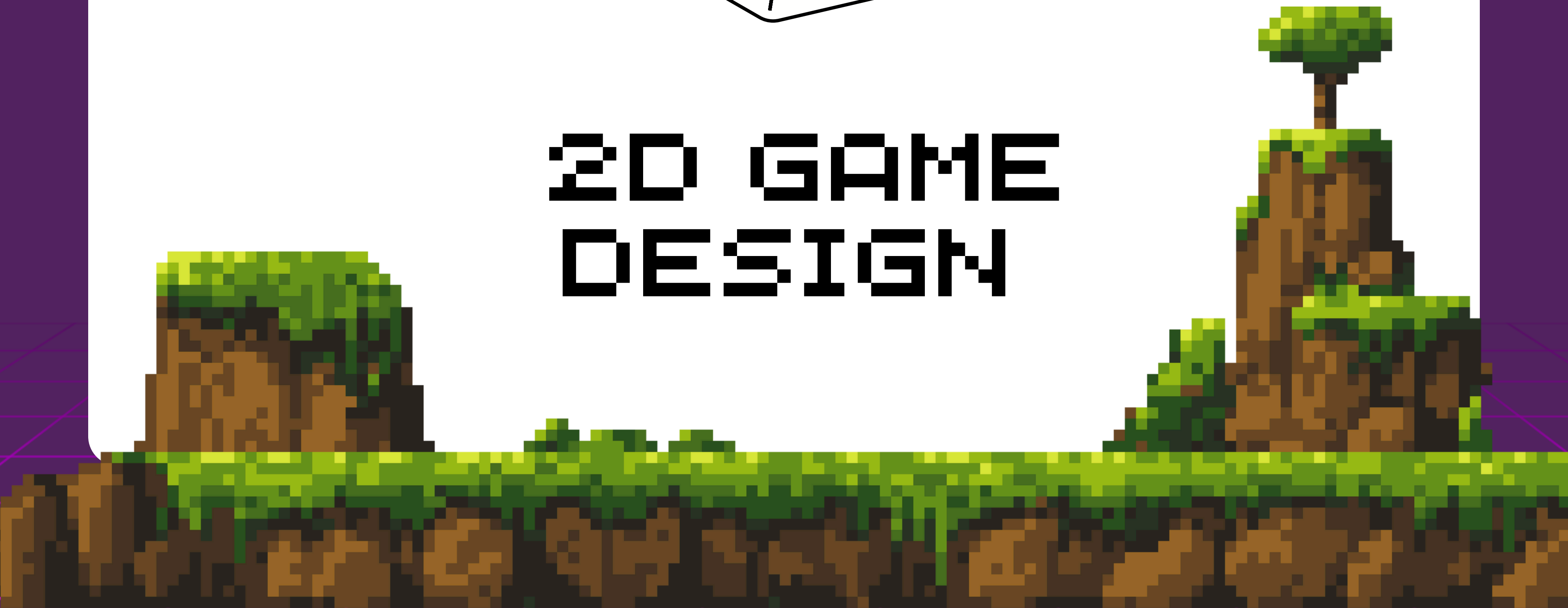


START





2D GAME DESIGN



2D GAME DESIGN

Concept & Planning

- *Before developing a 2D game, you must define the game's core idea, mechanics, target audience, and platform.*

Level Design

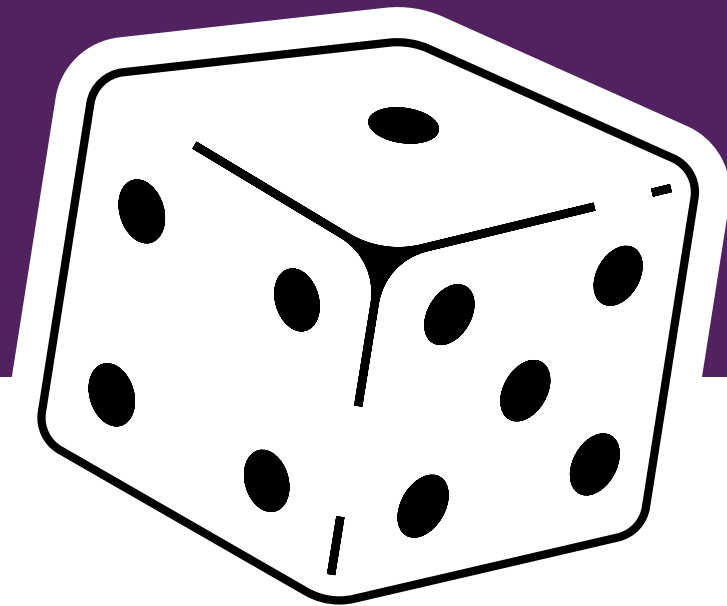
- *Creating engaging levels that provide challenges while maintaining balance and fairness for players.*

User Interface (UI)

- *Designing menus, buttons, HUDs (Heads-Up Displays), and other interactive elements for a smooth player experience.*

Game Mechanics

- *Defining how the player interacts with the game world, including movement, physics, and rules.*



SPRITES



SPRITES



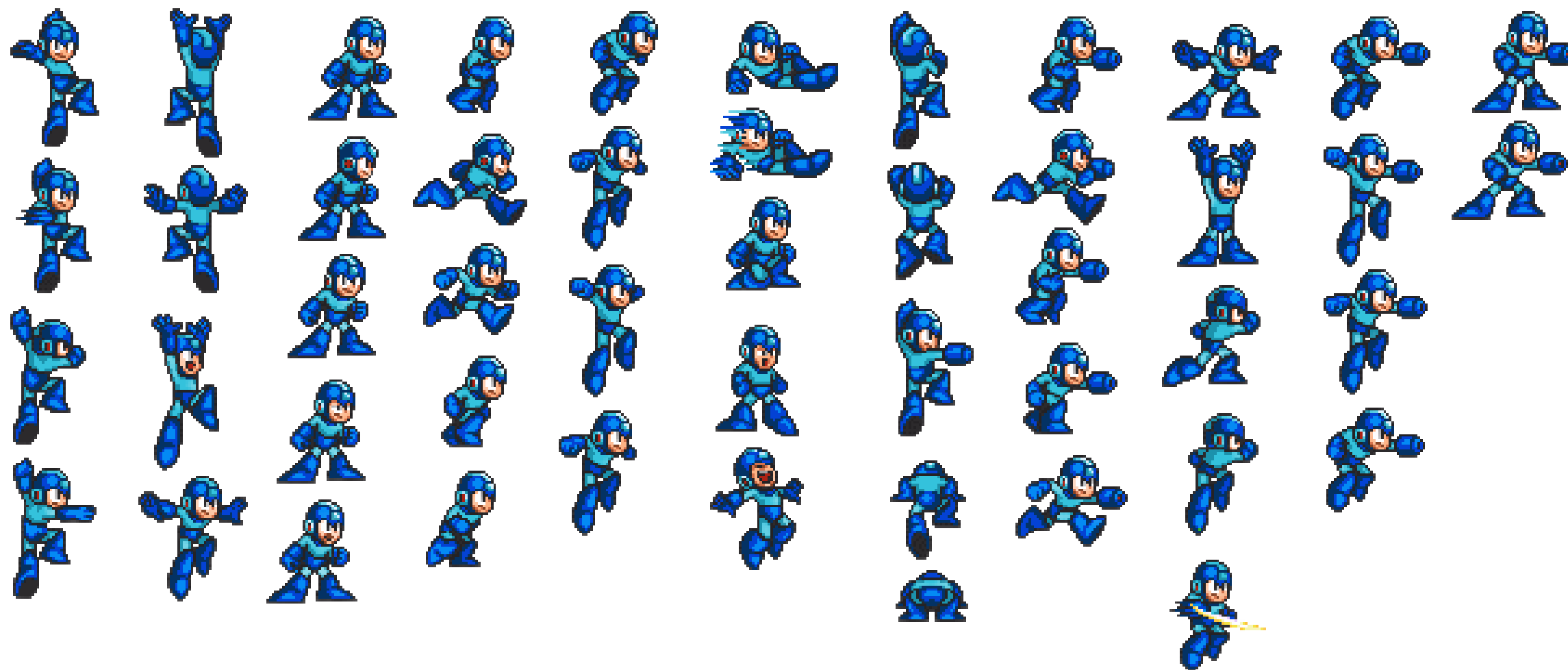
SPRITES

Definition

- *A sprite is a 2D image or animation used in a game to represent characters, objects, or effects.*

Sprite Sheets

- *A collection of images placed in a grid format, used for animations or different game states.*



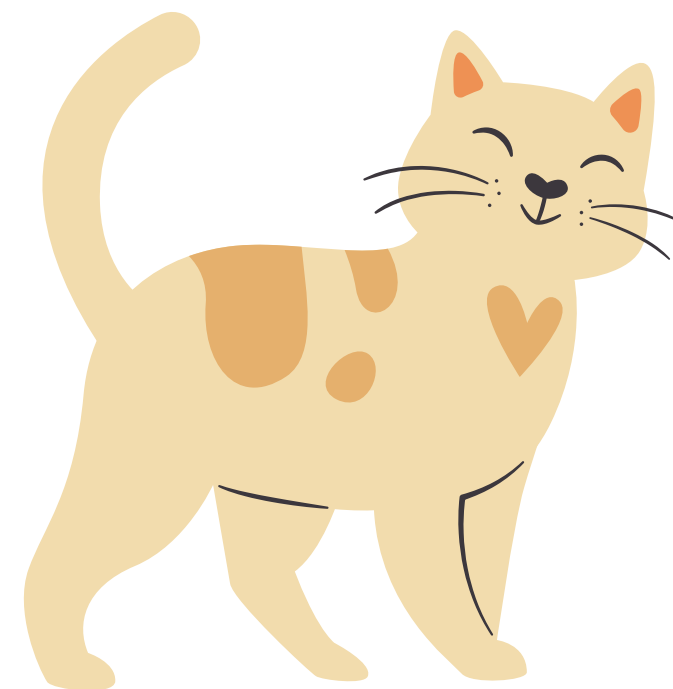
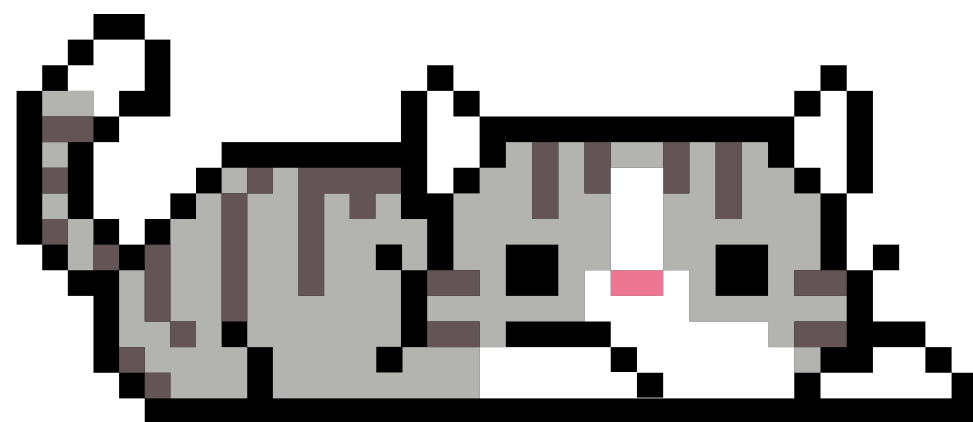
SPRITES

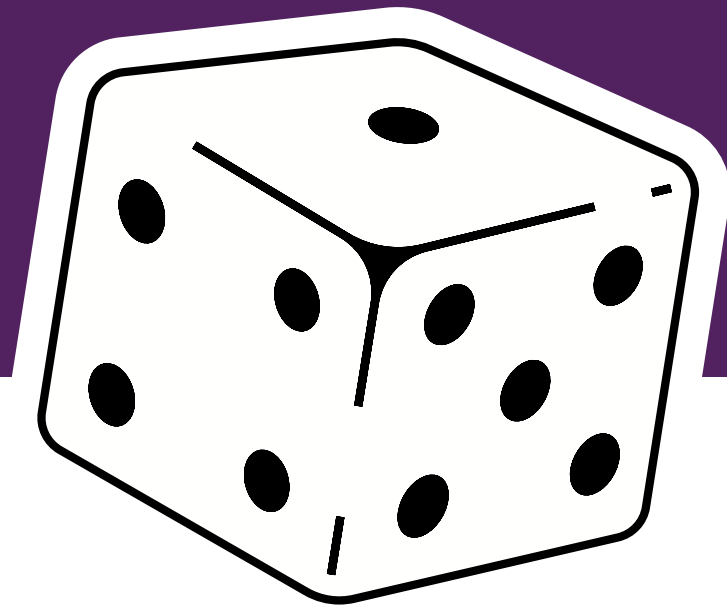
Transparency & Layers

- *Properly managing transparency (alpha channels) to ensure smooth rendering.*

Pixel Art vs. Vector Art

- *Pixel Art: Uses small pixels to create detailed, low-resolution graphics.*
- *Vector Art: Uses mathematical shapes to create scalable and smooth images.*





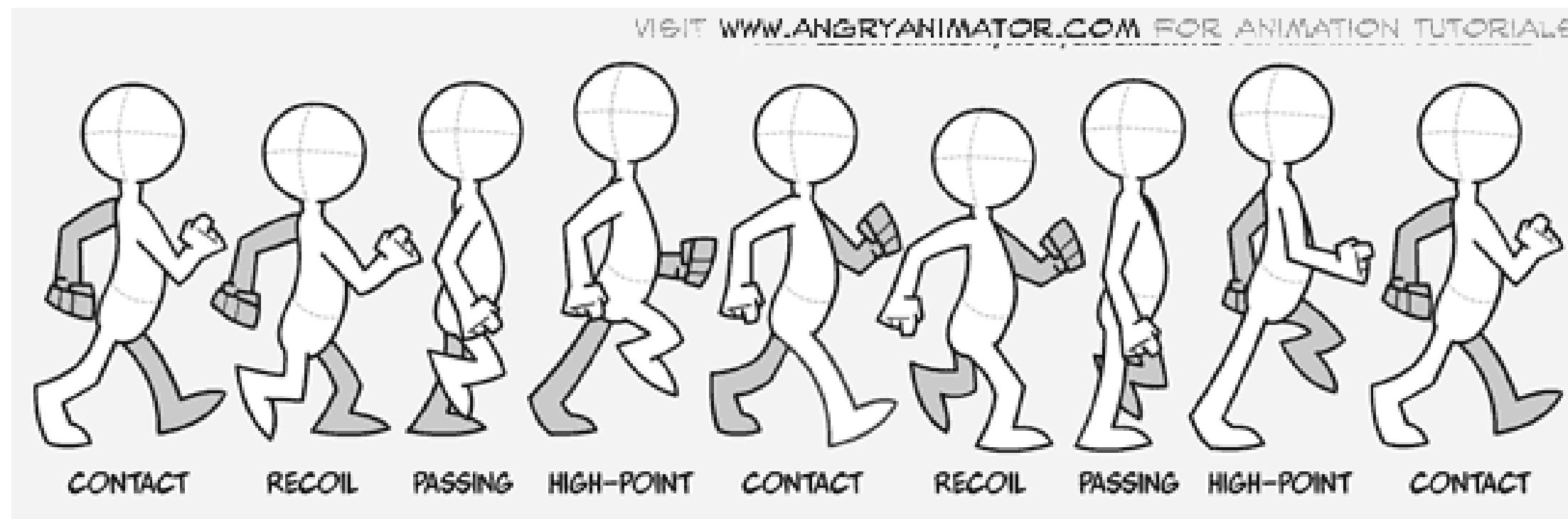
ANIMATION



ANIMATION

Frame-by-Frame Animation

- *Multiple images (frames) displayed in sequence to create movement.*



Sprite Animation:

- *Walking, running, jumping, attacking, etc.*
- *Use sprite sheets for efficient animation handling.*

ANIMATION

Tweening

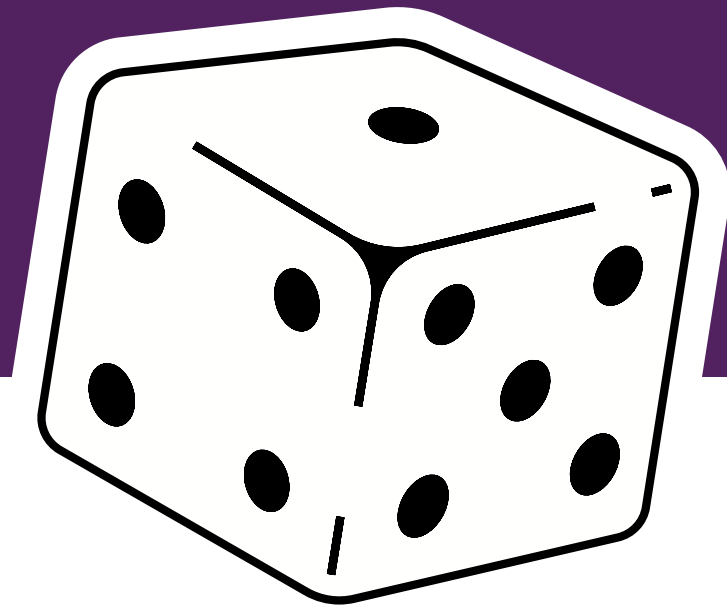
- *Tweening is an animation technique that creates a smooth transition between keyframes. It's also known as inbetweening.*

How Does Tweening Work?

- *Keyframes: Keyframes are the images at the beginning and end of a transition.*
- *Inbetweens: Inbetweens are the images that go between keyframes.*
- *Result: Tweening creates the illusion of movement by smoothly transitioning one image into another.*

Frame Rate (FPS)

- *The number of frames displayed per second; higher FPS results in smoother*



COLLISION DETECTION



COLLISION DETECTION

Definition

- *Detecting when objects in a game interact with each other.*

Types of Collision Detection

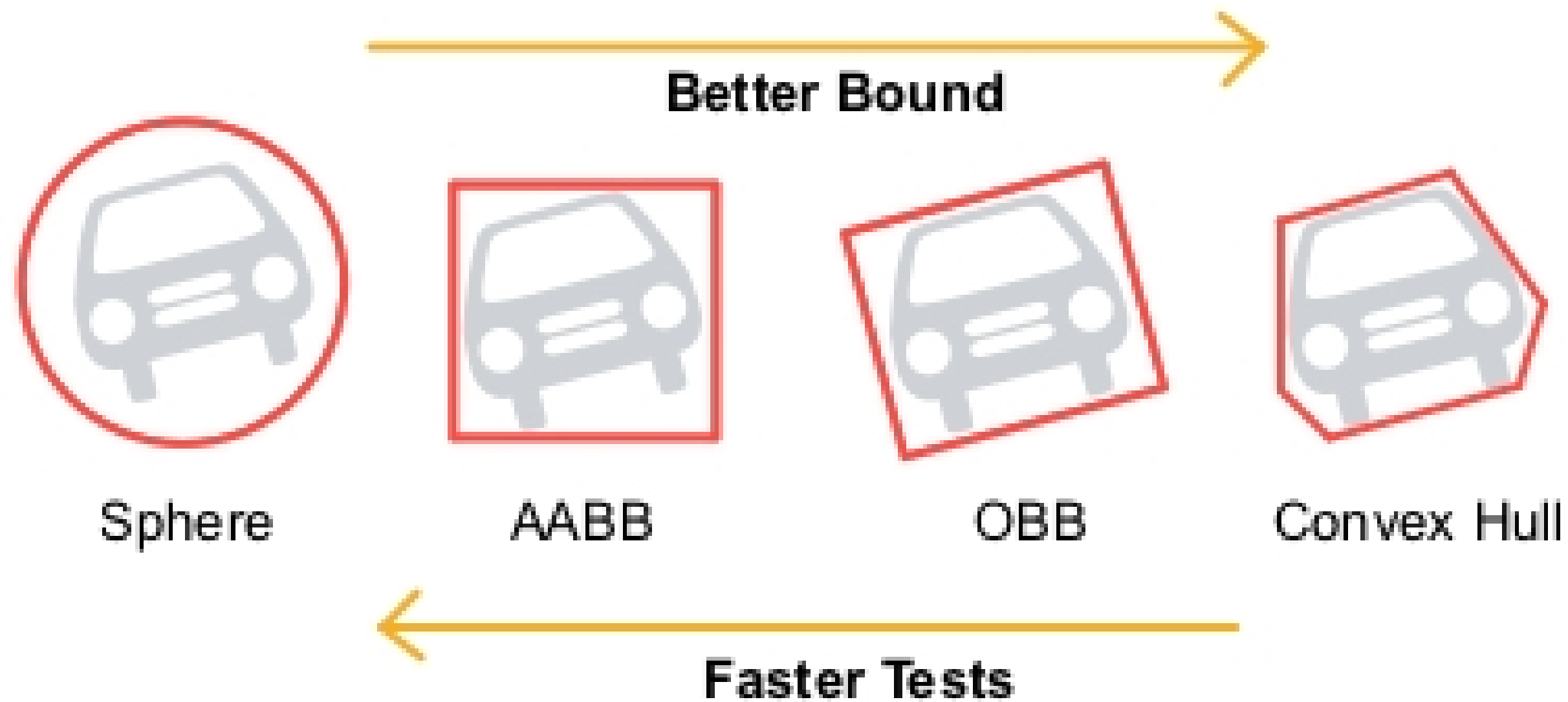
- *AABB (Axis-Aligned Bounding Box): Uses rectangles to detect overlapping objects.*
- *Circle Collision: Uses the distance between object centers for detection.*
- *Pixel-Perfect Collision: Checks pixel-level intersections for precise detection.*

Collision Responses

- *Preventing movement (e.g., walls blocking a player).*
- *Triggering events (e.g., collecting coins, taking damage).*
- *Bouncing or physics-based interactions.*

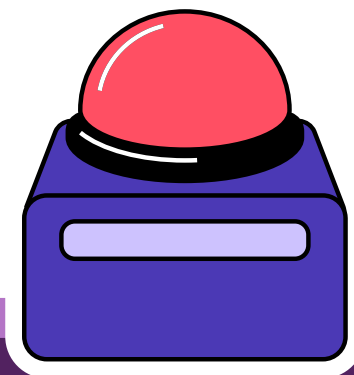
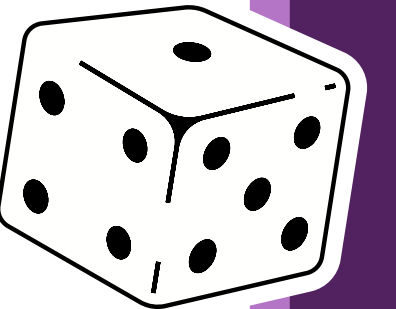
COLLISION DETECTION

Bounding Volumes



CONCLUSION

These fundamental concepts will help you understand and build a solid foundation for 2D game development.





THANK YOU
FOR PLAYING!

